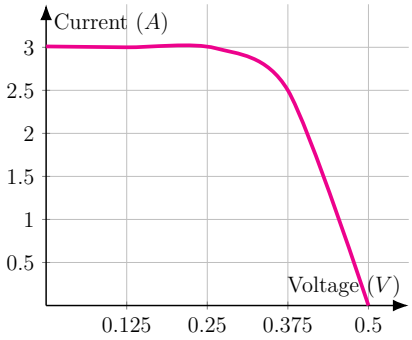
EE2015: Electric Circuits and Networks

Tutorial 2

(August 13, 2025)

1. The current-voltage characteristic of a silicon solar cell exposed to direct sunlight is shown on the right. It is obtained by placing different-sized resistors across the two terminals of the device and measuring the resulting currents and voltages.

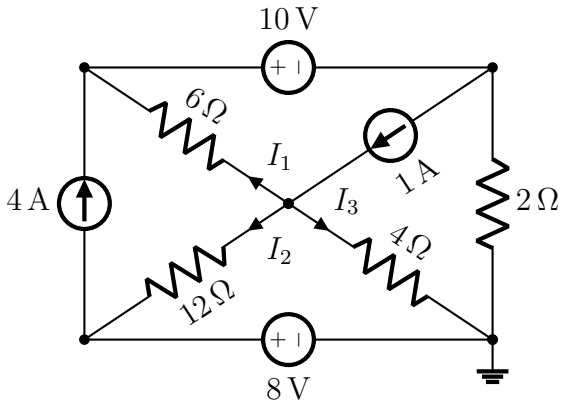


   (a) What is the value of the short-circuit current?

   (b) What is value of the voltage at open circuit?

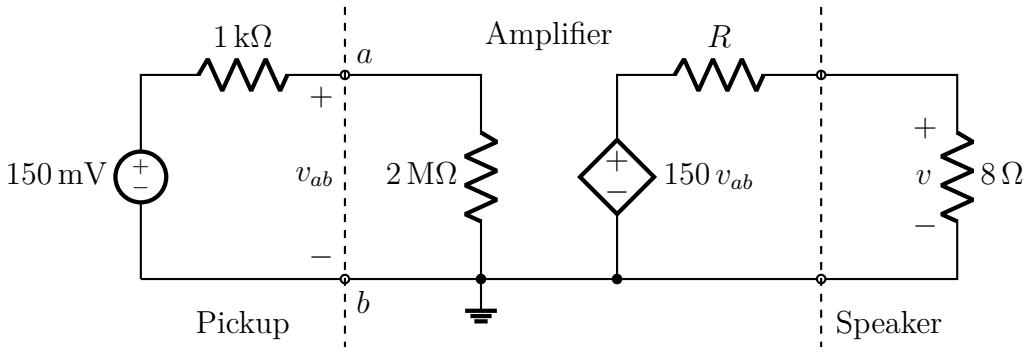
   (c) Estimate the maximum power that can be obtained from the device?

2.



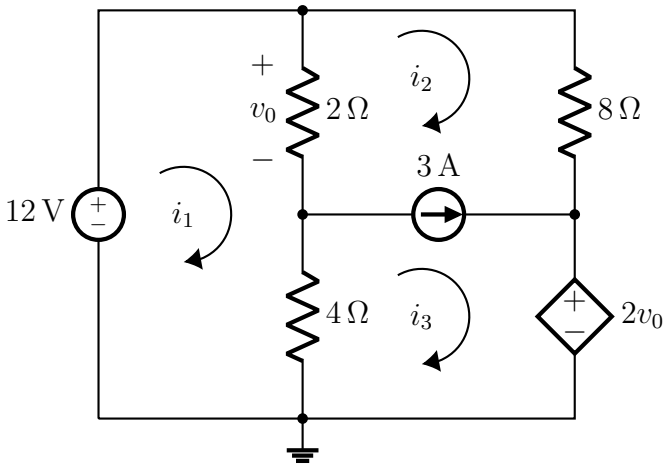
Consider the circuit shown on the left. What are the values of I_1, I_2, and I_3 ?

3. The equivalent circuit of a phonograph pickup, stereo amplifier, and speaker connected in series is as show below. Determine the resistance R so that the voltage v across the speaker is 16 V. What is the power delivered to the speaker?



4.

For the circuit shown on the right, find i_1, i_2, and i_3.



5.

Consider the circuit shown on the left. Find $i(t)$ for $t \geq 0$.

